

Final Project: Stacking

Please read the description to understand the story of the project. And we'll ask you to do your task in this project.

Story Description

In recent internet culture, "goblins" have become a humorous symbol of quirky, self-deprecating individuals who embrace their flaws. But what most people don't know is that goblins are not just memes—they are real. Hidden in a far-off mountain, Goblin Village thrives, and its residents are as peculiar as their online counterparts.

This year, the goblins were hosting their annual "Smartest Goblin Contest," a challenge that pushed the brightest goblins to solve the village's toughest problems. Five goblins stepped up, each bringing their unique skills to the table:

1. **Logic Goblin:** Precise and methodical, a firm believer in clear and rational solutions.
2. **Tree Goblin:** A systematic thinker who broke problems into smaller, manageable pieces.
3. **Naïve Goblin:** Cheerful and carefree, relying on simple assumptions to guide him.
4. **KnoW Goblin:** A social observer, making decisions based on the wisdom of those around him.
5. **Brainy Goblin:** A deep thinker, solving problems with his layered and intricate approach.

Each goblin worked hard, crafting their own solution to the challenge. However, their individual efforts fell short of perfection, and soon, the goblins were arguing

endlessly, each claiming their method was superior.

That's when the wise Goblin Elder appeared. "Why fight, young ones, when you can **stack**?" he asked with a sly grin.

"Stack?" the goblins repeated, puzzled.

"Yes, stack! Alone, you each have weaknesses, but together, your strengths can create something far greater. Goblins together, strong!"

Inspired by the elder's wisdom, the goblins put aside their differences and combined their efforts. Layer by layer, they stacked their methods, creating a solution far more powerful than any of their individual approaches.

The goblins cheered, repeating the elder's words: "Goblins together, strong!" From that day, the "Goblin Stacking Method" became a legend, proving that even goblins can achieve greatness when they stand united.

Your Job

Now, as a smart student at NYCU, you realize that machine learning models can stack together and become powerful too.

Thus, what you must do is to implement **Stacking** in this project.

Dataset Description

The data for this project is real-word data for hospitalizing patients. Each patient is described by 77 *features* (F1~F77), together with a binary outcome. The project involves a training model with provided datasets (train_X.csv & train_y.csv) and final testing on an independent test set (test_X.csv & test_y.csv).

Note:

*In this project, we only provide test_X.csv for you. This means that you cannot directly use the test data to evaluate the model's performance. Therefore, we recommend using **K-fold cross-validation** to decide the best model and **predict the label** for the testing data. We will use your predict label to help you*

calculate the final prediction score.

Language

You can implement your code by using either **Python3** or **C++**. (Just need to submit one of them)

Requirement

1. Download the necessary library and preprocess the data

This part is the same as the previous project.

2. Make sure you are implementing the base learner correctly

Because the performance of stacking is highly related to the result of base learner, please make sure you are implementing the base learner correctly.

We will provide a model test for you to test the model. The test is much easier than the dataset we use in this project, please make sure you can pass the test first.

3. Build your stacking model

We have written some start code framework, please finish your implementation of stacking according to it. Because stacking is highly flexible, you can build your stacking based on any kind of structure by **using the models implemented in previous projects** but notice your main idea of stacking should be correct. We will ask you to explain the idea of stacking in the report.

Note: You can also use Logistic Regression provided in project #1 to build stacking.

4. K-Fold Cross-Validation

In this project, you need to do **10-Fold** Cross-Validation on training data and show the result to us by **pasting your result in the report** like project #1 and #2. The result of K-Fold CV should be the average scoring of 10-fold, e.g. if you have a list stored scoring of 10-fold such as: [0.55, 0.54, 0.53, 0.6, 0.63, 0.45, 0.55, 0.7, 0.6, 0.58], then you should print out its mean value: 0.573.

*Note: The result of K-Fold CV will be in the part of your score in this project, please **remember** to paste it in your report!!!*

5. Predict the label of the testing data

Use your well-structured stacking model to predict the label of the independent testing data and upload it to E3.

6. Do your implement **from scratch**

You are not allowed to use any machine learning libraries to directly build stacking. That is, you are only allowed to use the model you wrote in previous projects to build your stacking model. If we find you're using some built-in functions to directly build any model, your score of this project will directly deduct 40%.

Grading Policy (Total 100 pt)

1. Report Spec (50%) (follow this format to write your report)

A. Explain the code detail of your stacking model (30%)

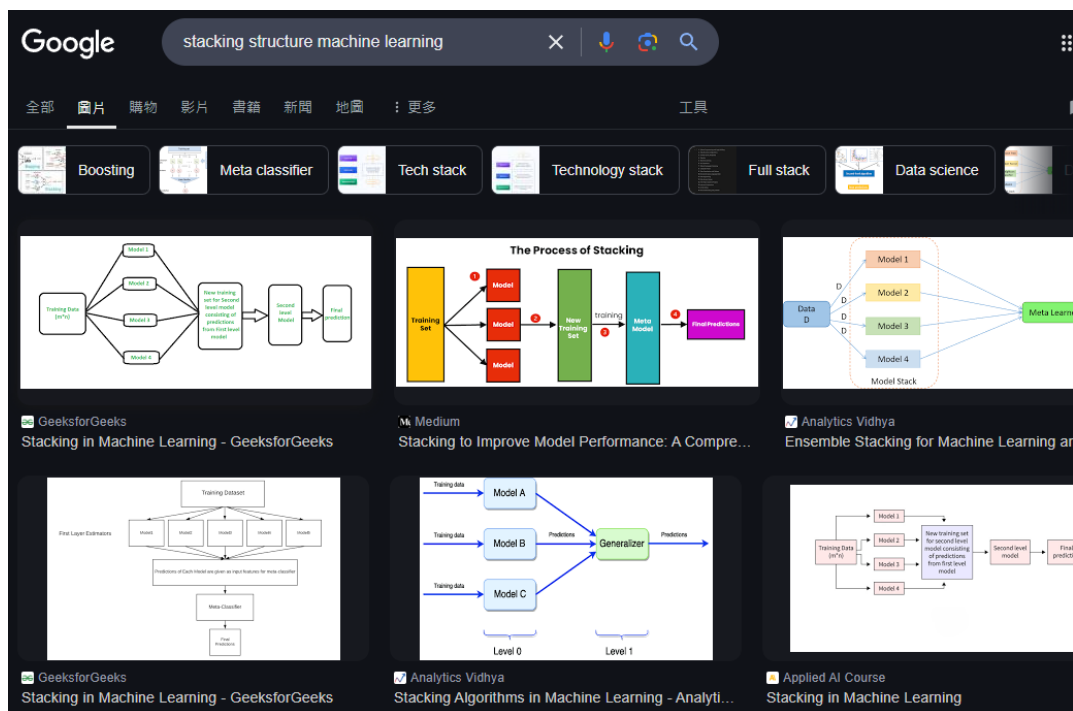
Note:

- This part should include workflow of your code, pseudocode or screenshot of code, and code explanation in detail.

B. Explain the concept of stacking and provide the structure (20%)

Note:

- Make sure the concept of stacking is correct, or you'll lose some points here.
- For the structure of stacking, you should provide an illustration such as:



C. Discussion (Optional)

Anything you want to share. (e.g. If the performance of your stacking model is worse than that of your base learners, you can discuss the possible causes.)

2. Model Test (20%)

Like we said before, the performance of stacking is highly related to the result of base learner. When grading the previous project, we found that some students may implement their model incorrectly, which resulted in not passing the base line of the previous projects. Thus, to make sure you implemented the models correctly, please pass the model test we provide for you first. The details of the model test will be shown in ***proj_final_guildline_python.pdf*** or ***proj_final_guildline_cpp.pdf***.

For the score in this part, you'll get:

- Total 10 pts if you pass the all the tests of best learner
- Total 10 pts if you pass the test for stacking

Grading Formula:

$$Base = \frac{\# \text{ base learner passed}}{5} \times 1$$

(e.g. if you pass 3 base learners, you will get: $3/5 * 1 = 0.6$)

$$Meta = \begin{cases} 1, & \text{if stacking test passed,} \\ 0, & \text{if stacking test failed} \end{cases}$$

$$Total_{\text{Model Test}} = (Base + Meta) \times 10$$

(e.g. $Total_{\text{Model Test}} = (0.6 + 1) * 10 = 16$)

3. Performance (30%)

- For your result of performance, we will consider both the scoring of K-Fold Cross-Validation (10-fold) on training set (60%) and the pred label that you submit to E3 (40%)
- Scoring = $0.3 * \text{acc} + 0.35 * \text{f1 score} + 0.35 * \text{mcc}$
- PR Score:

The Performance Rating (PR) Score is determined by the rank of your final scoring among all students, which considers the scoring of K-Fold Cross-Validation and the predicted labels.

- 30/30, if $\text{PR} > 90$
- 27/30, if $90 > \text{PR} \geq 85$
- 25.5/30, if $85 > \text{PR} \geq 80$
- 24/30, if $80 > \text{PR} \geq 75$
- 22.5/30, if $75 > \text{PR} \geq 70$
- Below PR 70: Scores decrease progressively, e.g. PR 70~65 may score 18/30 points.

Performance Metric	Definition
Recall ^a	$\text{TP}/(\text{TP} + \text{FN})$
Precision ^b	$\text{TP}/(\text{TP} + \text{FP})$
ACC	$(\text{TP} + \text{TN})/(\text{TP} + \text{TN} + \text{FP} + \text{FN})$
F1-score	$\frac{2 \times \text{Recall} \times \text{Precision}}{\text{Recall} + \text{Precision}}$
MCC	$\frac{\text{TP} \times \text{TN} - \text{FP} \times \text{FN}}{\sqrt{(\text{TP} + \text{FP}) \times (\text{TP} + \text{FN}) \times (\text{TN} + \text{FP}) \times (\text{TN} + \text{FN})}}$
AUC	Area under the ROC curve

TP, true positive; TN, true negative; FP, false positive; FN, false negative; MCC, Matthews correlation coefficient; ACC, accuracy; AUC, area under the curve; ROC, receiver operating characteristic.

^a Recall is equivalent to sensitivity in its definition.

^b Precision is equivalent to positive predictive value in its definition.

Note: We will use the scoring you of K-Fold CV pasted in the report and the predict label you upload to E3 to calculate your final PR value. Also, be sure

you **set up the random seed** of your best model to avoid controversy.

e.g.

```
import random
random.seed(0)
```

```
import numpy as np
np.random.seed(0)
```

Submission Deadline:

Deadline: 12/28 (Sat.) 23.59. p.m.

Submission format: (Remember to upload your predict label)

- You need to submit **all your code in one folder** together with your **report file, prediction label**, and zip them into one **.zip** file.
- Name your predict label as “*pred_final_<yourStudentID>_<yourName>.csv*”
e.g. “*pred_final_312551120_陳柏憲.csv*”
- Name your code folder as “*<language>_final_<yourStudentID>_<yourName>*”,
e.g. “*C++_final_312551120_陳柏憲*” or “*Python_final_312551120_陳柏憲*”
- Name your report file as “*report_final_<yourStudentID>_<yourName>.pdf*”,
e.g. “*Report_final_312551120_陳柏憲.pdf*”
- Name your .zip file as “*ml_final_<yourStudentID>_<yourName>.zip*”,
e.g. “*ML_final_312551120_陳柏憲.zip*”

(Uppercase or lowercase of the letters is fine)

Punishment Policy:

Except for what we mentioned in this file, please refer to *[project_policy.pdf](#)* we provided in project #1.

We will also run your code to reconfirm the result, if we **cannot compile and run** your code, we will ask you to re-submit your code, and you will **lose some points** in the final score of this project. (You can tell us how to execute your code and where we should modify the path as the previous project do)

Note:

*We will randomly select 10 students for **oral demonstration** after the deadline of the final project. We will announce the details of the oral demonstration as soon as possible.*

Good luck for your final project~

TA